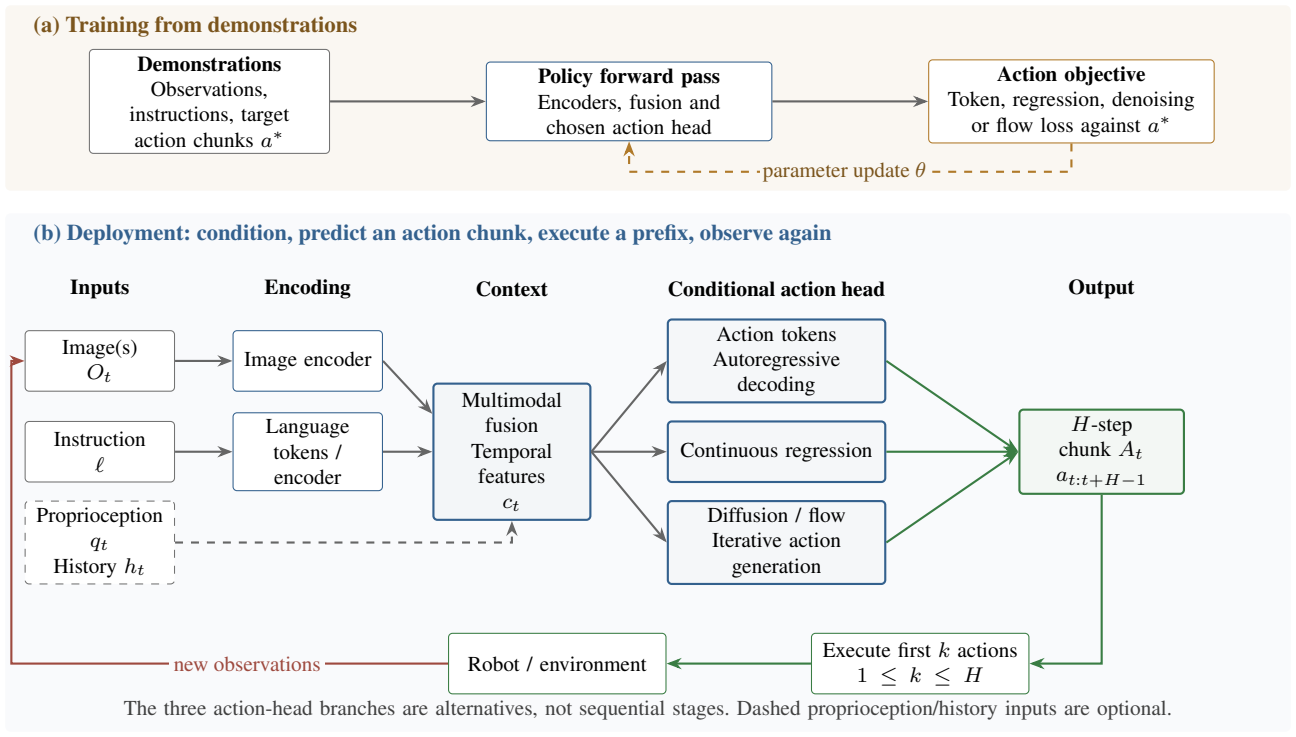

(a) Training from demonstrations
Demonstrations
Observations, instructions, target action chunks a^*
Policy forward pass
Encoders, fusion and chosen action head
Action objective
Token, regression, denoising or flow loss against a^*
parameter update θ
(b) Deployment: condition, predict an action chunk, execute a prefix, observe again
Inputs
Encoding
Context
Conditional action head
Output
Image(s) O_t
Instruction ℓ
Proprioception q_t History h_t
Image encoder
Language tokens / encoder
Multimodal fusion Temporal features c_t
Action tokens Autoregressive decoding
Continuous regression
Diffusion / flow Iterative action generation
H-step chunk A_t $a_{t:t+H-1}$
Execute first k actions $1 \le k \le H$
Robot / environment
new observations
The three action-head branches are alternatives, not sequential stages. Dashed proprioception/history inputs are optional.